UPDATED LESSON PLAN: “Why a Big Idea Isn’t Enough”

Grade: MYP 1
Duration: 40 minutes
Purpose: Students discover the role of KC, RC, and GC by experiencing the limitations of using only a big idea.

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1. Learning Objectives

By the end of the lesson, students will be able to:

1. Recognize that a single big idea (KC) is too broad unless narrowed.
2. Understand the need for subject-specific ideas (RC).
3. Explain why learning must be grounded in real-world contexts (GC).
4. Build a simple mini-unit with KC–RC–GC.

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2. Materials Required

- Chart paper or A4 sheets
- Markers
- Printed or projected big idea options
- Exit ticket slips

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3. Lesson Sequence (40 minutes)

PHASE 1: Introduction & Challenge (5 minutes)

Teacher Script

“Today you will plan a lesson using only one word. You will be the teacher.
These are your choices:”

Write or display:

1. Food
2. Animals
3. Weather
4. Games
5. Water
6. Home

“Choose one word. This is the entire focus of your Grade 6 lesson.”

“Your group must list 5–7 things you would teach about your chosen word.”

No explanation or guidance yet.

PHASE 2: Group Planning – Only the Big Idea (10 minutes)

Teacher Actions

- Circulate, observe, do not guide.
- Repeat: “Use your chosen word. Anything connected to it is okay.”

Student Actions

Groups brainstorm freely.

Expected outcomes:

- **Food:** nutrition, recipes, shopping, prices, farming, festivals
- **Animals:** habitats, stories, food chains, pets, claws, classification
- **Weather:** seasons, disasters, tools, data, clouds, clothing
- **Games:** rules, scoring, teamwork, maths, culture, strategy
- **Water:** states, rivers, pollution, uses, safety, conservation
- **Home:** family, chores, materials, design, safety, culture

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PHASE 3: Sharing & Problem Identification (5 minutes)

Teacher Script

“Let’s hear each group’s teaching points.”

After presentations:

“What do you notice about the topics?
Are they all from one subject?
Are they very wide?
Would it be easy to teach everything in one lesson?”

Students realize the big idea is too broad.

PHASE 4: Introducing the Subject Idea (RC) (8 minutes)

Teacher Script

“To make a unit teachable, we need a **subject idea**.
This is a smaller idea that belongs to a single subject.”

Provide sample subject ideas:

Food

- Nutrition (Science)
- Prices (Math)
- Culture (I&S)
- Stories (Language)

Animals

- Habitats (Science)
- Characters (Language)
- Pets & society (I&S)

Weather

- Measurement (Math)
- Seasons (I&S)
- Weather tools (Science)

Games

- Rules (Design)
- Strategy (Math)
- Teamwork (PHE)
- Stories (Language)

Water

- States (Science)

- Conservation (I&S)
- Measurement (Math)

Home

- Materials (Design)
- Family structures (I&S)
- Safety (PHE)

“Choose ONE subject idea and rewrite your teaching points.”

PHASE 5: Real-World Lens (GC) (5 minutes)

Teacher Script

“Learning also needs a real-world lens—a place or situation where this happens.”

Give kid-friendly lenses:

- Home
- School
- Community
- Environment
- Health
- Culture
- Technology

“Pick one and make your teaching points more connected and meaningful.”

PHASE 6: Reflection & Reveal (2 minutes)

Teacher Script

“What changed when you added the subject idea?
What changed when you added the real-world lens?
Which version was easier to teach?”

Then reveal:

“In MYP:

- Big idea = Key Concept
- Subject idea = Related Concept
- Real-world lens = Global Context.”

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4. Exit Ticket (5 minutes)

Exit Ticket – Write 1–2 sentences each:

1. Why is a big idea alone not enough for planning a unit?
2. How does a subject idea help?
3. Why do we need a real-world lens?
4. Which of the three (big idea, subject idea, real-world lens) helped you the most today and why?

Collect before students leave.

Strand	Activity	Goal	Deliverable	ATL Focus	Week
A.i	Why lip balm? Explore personal needs and sustainability	Identify a design need based on personal experience	Reflection paragraph on lip balm need	Communication	Week 1
A.ii	Research natural ingredients, packaging, and trends	Collect and organize relevant primary/secondary research	Research worksheet and ingredient analysis table	Media literacy	Weeks 2–3
A.iii	Analyze 3–5 existing products using images or samples	Learn what makes a good product through comparison	Product analysis table and annotated examples	Media literacy	Week 4
A.iv	Draft a design brief for your lip balm	Summarize target audience and goals	Written design brief (template)	Communication	Week 5

Strand	Activity	Goal	Deliverable	ATL Focus	Week
B.i	Write a design specification (color, scent, packaging etc.)	Create clear criteria for your product	Specification chart with icons	Organization	Week 6
B.ii	Sketch 2–3 lip balm design ideas (label & container)	Generate creative and feasible ideas	Sketches with labels and annotations	Creative thinking	Week 7
B.iii	Choose your best design and justify the choice	Select a final idea based on reasoning	Annotated final design with rationale	Decision-making	Week 8
B.iv	Create diagrams/recipe for lip balm production	Plan making process clearly	Ingredient list, steps, tools, and safety	Time management	Week 9

Strand	Activity	Goal	Deliverable	ATL Focus	Week
C.i	Create a production checklist and time plan	Organize production process and materials	Action plan with timeline	Organization	Week 10
C.ii	Make the lip balm (teacher demo → student practice)	Apply technical skills safely and correctly	Final product and photos	Collaboration	Week 11
C.iii	Follow and revise your plan during making	Adapt plan as needed during execution	Marked-up plan or logbook	Reflection	Week 12
C.iv	Reflect on what changed during the process	Explain changes and their reasons	Post-production reflection sheet	Critical thinking	Week 13

Strand	Activity	Goal	Deliverable	ATL Focus	Week
D.i	Create testing methods	Create Review sheet	Checklist aligned with design specification	Reflection	Week 14
D.ii	Evaluate how well it meets your specifications	Measure product success	Specification checklist reflection	Evaluation	Week 14
D.iii	Describe what you would improve	Identify ways to improve design	Written reflection	Growth mindset	Week 15
D.iv	Consider environmental and social impact	Reflect on sustainability and self-use	Poster or paragraph: “Why my balm is eco-friendly”	Media literacy	Week 15

Why Lip Balm?

Understanding Personal Needs, Sustainability, and Ethical Design

What Is Lip Balm?

Lip balm is a soft, waxy product used to protect and moisturize the lips, which are particularly sensitive and prone to dryness because they lack oil glands. Unlike other areas of your skin that naturally stay moisturized due to sebum (natural oils), your lips dry out quickly in hot, cold, windy, or dry weather. They are also exposed to food, drinks, UV rays, and pollution, making protection even more important. That's where lip balm comes in—it forms a barrier and helps seal in moisture.

Most commercial lip balms are made from a mix of waxes, oils or butters, and additives like fragrances and colorants. While they may feel good at first, not all of them are actually healing. Some may contain ingredients that only give temporary relief, leaving your lips drier over time and making you reapply again and again.

Common Lip Balm Problems

It's important to understand that not all lip balms are good for you—or the planet. Some store-bought lip balms may contain:

- **Petroleum-based waxes**, like petrolatum or paraffin, which are byproducts of crude oil and are not biodegradable.
- **Synthetic fragrances** that may cause allergies or skin sensitivity.
- **Artificial preservatives** like parabens, which have raised health concerns.
- **Plastic packaging** that cannot be easily recycled.

These products may look shiny and smell sweet, but many of them come with hidden environmental and health costs. In fact, some lip balms contain ingredients that can actually dry out your lips, encouraging you to use more frequently and buy more often.

Why Make a Natural Lip Balm?

Making your own natural lip balm allows you to take control of what goes on your skin. You can select ingredients that are safer, more effective, and more environmentally responsible. A natural, sustainable lip balm is:

- **Plant-based**, using oils and butters that are biodegradable and gentle on skin.
- **Free of harsh chemicals**, artificial fragrances, and synthetic preservatives.
- **Packaged sustainably**, often in metal tins, glass jars, or biodegradable paper tubes.
- **Customizable**, letting you create a balm that suits your exact needs, such as texture, scent, and color.

For example, if you like citrus scents, you might add orange essential oil. If you want a tint, you could use beetroot powder. And if you prefer a firmer balm for warmer climates, you can adjust the wax ratio.

Common Ingredients in Natural Lip Balms

Here's a breakdown of the types of ingredients you'll be using when designing your own lip balm:

1. Waxes

Waxes give lip balm structure and help it hold shape. They also create a protective layer over your lips.

- **Beeswax** – Common in natural lip balms. Moisturizes and protects. However, not vegan.
- **Candelilla Wax** – Plant-based, a great vegan alternative to beeswax.
- **Carnauba Wax** – A harder wax from the leaves of the carnauba palm; also vegan.

2. Butters

These are rich, creamy ingredients that deeply moisturize.

- **Shea Butter** – A popular choice for hydration; rich in vitamins.
- **Cocoa Butter** – Adds a chocolaty scent and smooth texture.

- **Mango Butter** – Lighter, non-greasy, and great for sensitive skin.

3. Oils

Oils make the balm easy to spread and deliver nutrients to the skin.

- **Coconut Oil** – Has antimicrobial properties and a lovely tropical scent.
- **Sweet Almond Oil** – Lightweight and rich in Vitamin E.
- **Jojoba Oil** – Closely resembles the skin's natural oil; non-greasy.

4. Additives (Optional)

- **Essential Oils** – For fragrance and sometimes skin benefits (like lavender for calming).
- **Natural Colorants** – Beetroot powder, cocoa powder, turmeric (use with care!).
- **Vitamin E Oil** – A natural preservative and skin conditioner.

🧘🏾‍♀️ Ethical and Sustainable Considerations

Being an ethical cosmetic designer means thinking about **how your choices affect people and the planet**. Here are some issues to be aware of:

🔹 **Packaging Waste**

Most store-bought lip balms are sold in small plastic tubes. Over **200 million tubes** are discarded yearly. These are often too small to recycle and end up polluting landfills or oceans. Alternatives include:

- Reusable metal tins
- Glass jars
- Compostable paper tubes

🔹 **Ingredient Sourcing**

Some ingredients may be "natural" but come with ethical concerns. One such ingredient is **mica**, used to add shimmer or tint in cosmetics. Mica is mined in several countries, including India, where reports of child labor and unsafe mining conditions have been well documented. As designers, we must research ingredients and look for **ethically sourced mica** or avoid it

altogether. Choose suppliers that are transparent about where and how their ingredients are obtained.

🔹 **Animal Testing**

Many mainstream cosmetics brands still test their products on animals. As young consumers and creators, you can support cruelty-free practices by ensuring your product is **not tested on animals** and contains **vegan ingredients**, if aligned with your values.

🔹 **Carbon Footprint**

Try to use **local ingredients** or those with lower transport emissions. For instance, if shea butter is locally available, it's a more eco-friendly choice than imported cocoa butter.

Why This Matters to YOU

When you create a lip balm for yourself, you're not just making something to look pretty or smell good. You're designing a product based on:

- Your **personal needs** (dry lips, allergies, scent preferences)
- Your **values** (sustainability, ethical sourcing, cruelty-free products)
- Your **creativity** (texture, color, shape, packaging)

This process helps you become a **more thoughtful consumer** and a **more responsible designer**. You will learn how small decisions like using paper packaging or skipping synthetic fragrance can make a **big impact**.

Stage	What You Do	What Students Do
1. Starter Discussion (10 mins)	Prompt: “Why do people use lip balm?” / “What annoys you about store-bought ones?”. Select your favourite lip balm and Make a list of the ingredients.	Share personal experiences in small groups
2. Provide a Simple Info Pack	Give them a **2-page doc** with key info, including:	

- What lip balm is made of
- Common issues (e.g. dryness, allergies, plastic waste)
- Sustainability and natural alternatives

🔍 STEP 1: Read & Highlight

Read the information sheet titled **“Why Make Your Own Lip Balm?”**

- Highlight or underline:
    - Problems with commercial lip balms
    - Key ingredients and their functions
    - Sustainable and ethical practices
    - Packaging types and environmental impacts

🧠 STEP 2: Create a Mind Map

Based on what you read, organize your ideas using the **Mind Map Template** on the next page. You may draw it by hand or recreate it digitally.

Your mind map should have **4 branches**:

1. **Problems with Store-Bought Lip Balms**
    - Harmful ingredients

- Packaging issues
- Allergic reactions
- Reapplying often

2. **My Personal Lip Care Needs**
    - Dry lips? Allergies? Preferences?
    - What's missing in the current products I use?
3. **Sustainable Ingredients & Packaging**
    - Which natural ingredients do I like?
    - What kind of containers are eco-friendly?
4. **Ethical & Global Concerns**
    - Where do ingredients come from?
    - Any concerns about sourcing (e.g., mica)?
    - How can I support global sustainability?

🧪 STEP 3: Research Your Current Lip Balm

Use your lip balm or pick one you have used before.

1. Search its **ingredients** online.
2. Look up:
    - What each ingredient does
    - Whether it is **natural or synthetic**
    - Whether it is **safe or potentially harmful**
3. Ask: Are there any ethical or environmental concerns about these ingredients?

Write 3–4 bullet points under your mind map noting:

- What did you discover about your lip balm?
- Do you want to use it again? Why or why not?

✍️ STEP 4: Reflection Prompt

Use this reflection sentence starter to summarize your findings:

> *“After exploring the ingredients in my current lip balm and reading about natural alternatives, I realized that...”*

💡 ATL Skills in Action:

- **Communication:** Summarizing, organizing, and visually representing information
- **Media Literacy:** Investigating how product ingredients and packaging affect health and the environment

Stage	Activity	Goal	ATL Skill	Deliverable	Time
🧠 1️⃣	**Hook Activity: The "Wrong" Lip Balm Game**Show 2–3 funny "failed" lip balm products (e.g. a cactus flavor, one in an impossible-to-open metal tin, or one that melts immediately in sun). Ask: "What went wrong?"Students discuss in groups and write what the designer *didn't think about*.	Realize the importance of identifying and articulating a real need.	Media literacy: Understanding product-user mismatch	Group notes: "What went wrong?" chart	15 mins
💬 2️⃣	**Why Do Designers Define Problems?**Class brainstorm (anchor chart or Padlet):"What happens if you skip defining the problem?"Then introduce 2 quick examples: a well-designed and a poorly-designed product.	Build understanding of the value of problem framing in design.	Communication	Class anchor chart: "Why Problem Statements Matter"	10 mins
✍️ 3️⃣	**Mini Workshop: Writing a Bold Problem Statement**Introduce a scaffold:"I need a lip balm that... because..."Guide them to include:– their lip needs– issues with current products– sustainability/ethical goals	Build writing confidence and structure thinking.	Communication	Problem statement draft (scaffolded)	25 mins
🔁 4️⃣	**Peer Swap & Suggest**Partner activity where students exchange their problem statements and offer 2 suggestions:– What's clear?– What could be stronger?	Practice giving and receiving actionable feedback.	Collaboration	Peer feedback slips	15 mins

Problem Statement Scaffold Worksheet

Problem Statement Scaffold

Goal: Write a clear, specific, and justified design problem statement for your lip balm project.

Use the sentence starters below to help organize your ideas. Complete all parts with details from your research and personal experience.

I need a lip balm that...
(Describe what your lip balm should do or who it should help — e.g., moisturize dry lips, be allergy-friendly, eco-friendly)

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because...
(Explain the problem or issues with current lip balms you have discovered — e.g., they cause dryness, use plastic packaging, contain harsh chemicals)

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and it should be...
(Add your sustainability or ethical considerations — e.g., made with natural ingredients, use recyclable containers, sourced fairly)

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so that...
(Describe the positive impact your lip balm will have on the user and/or environment)

Example:
I need a lip balm that gently moisturizes sensitive lips because many store-bought balms cause irritation and are packaged in plastic tubes. It should be made with natural, ethically sourced ingredients and packaged in reusable tins so that it is safe for me and better for the planet.

Slide Deck Outline for Lip Balm Design Fails Hook

Slide 1: Title Slide

Why Do Designers Need a Problem Statement?

Slide 2: Failed Product Example #1

Image of a funny or impractical lip balm (e.g., cactus-flavored, too hard to open)
Question: What do you think went wrong here?

Slide 3: Failed Product Example #2

Image of a lip balm that melts immediately or a bizarre container
Question: Why would this be frustrating for users?

Slide 4: Group Discussion Prompt

In your groups, discuss:

- What problems might the designer have overlooked?
- How could a clear problem statement have helped?

Slide 5: What Happens When We Skip Defining the Problem?

- Wasted resources
- Unhappy users
- Environmental harm
- Missed opportunities to innovate

Slide 6: Well-Designed Product Example

Image of a simple, eco-friendly lip balm with clear benefits
Notice: How clearly does this product solve a problem?

Slide 7: Conclusion

A strong problem statement helps designers:

- Understand the user's needs
- Address real issues
- Create better, more sustainable products

Explicit teaching : Research Question framing skills

PHASE 1: INQUIRY (20 minutes)

1. Question Sorting (10 min)

Students sort pre-written lip balm questions into groups **without labels**.

Prompt:

> “Which questions feel similar? Why?” Use this Slide

Students name groups in their own words.

2. Introduce the Matrix (10 min)

Reveal the **blank 3×3 grid** (no labels yet).

Ask:

- “Which group of questions could fit here?”
- “Which box feels easiest? Which is hardest?”

Only after discussion, reveal:

- RESOURCE / USER / PLANET
- FACT / REASON / CHOICE

PHASE 2: ACTION (40 minutes)

3. Explicit Teaching (10 min)

Teacher script (simple, clear):

> “Designers ask questions about:
>
> - the product

- the people
- the planet
  And they ask them in different ways:
- to find facts
- to understand reasons
- to make choices”

They also ask questions in **three ways**:

- **FACT** – to find information
- **REASON** – to understand why or how
- **DECISION** – to make a design choice

Connect explicitly:

- FACT = factual questions
- REASON = conceptual questions
- CHOICE = debatable questions

4. Guided Practice (15 min)

Students work in pairs.

Task:

Write **one question per box** related to their lip balm

Minimum:

- 1 FACT
- 1 REASON

- 1 CHOICE

Teacher circulates and prompts:

- “Who is this about?”
- “Will this help your design?”
- “Can this be answered in one line?”

5. Upgrading Questions (15 min)

Students:

- Take **one FACT question**
- Move it rightward:
    - FACT → REASON
    - or REASON → CHOICE

Example:

- FACT: *What ingredients are in lip balm?*
- REASON: *Why are these ingredients used for teenagers?*
- CHOICE: *Should teenagers avoid certain ingredients?*

Students write:

> “My upgraded question is better because…”

PHASE 3: REFLECTION (20 minutes)

6. Visible Thinking: I Used to Think / Now I Think (10 min)

Prompt:

- I used to think research questions were…

- Now I think research questions are…

Encourage reference to PRODUCT / USER / PLANET.

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7. Formative Assessment

Use the question matrix and create 4 questions for each box.

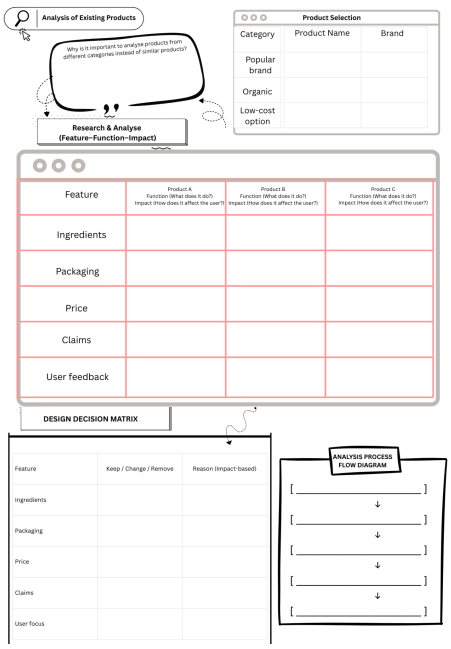
8. Exit Ticket (10 min)

Students complete:

> "My strongest research question is about the **___** and helps me make a **design decision** because…"

Product Research & SWOT Analysis Activity

[Tasksheet attached]



Step 1: Choose 3 Lip Balm Products

- Pick lip balms you can find online or in stores (or that you already have).
- Choose a mix, e.g., popular brand, natural/organic product, and a low-cost option.

Step 2: Research Each Product

Look for:

- Ingredients list
- Packaging type
- Price and availability
- User reviews or comments (if available)

- Claims (e.g., moisturizing, organic, eco-friendly)

Step 3: Complete the SWOT Analysis for Each Product

SWOT Category	Questions to Think About
Strengths	What is good about this lip balm? What do people like?
Weaknesses	What is not so good? What problems does it have?
Opportunities	How can this lip balm get better? What could be added?
Threats	What might make people stop buying it? What other lip balms compete with it?

Personal Persona & Design Brief Template

Step 1: Create Your Personal Persona

(Describe yourself as the user of your lip balm)

Field	Your Answer / Reflection
Name or Nickname	
Age	
Daily Activities / Lifestyle	What do you do most days? (e.g., sports, study, outdoor time)
Lip Care Habits	How often do you use lip balm? What do you like or dislike about the lip balms you've used?
Skin Type & Preferences	Do you have dry, sensitive, or normal lips? Do you prefer scented or unscented?
What Matters to You?	What's important to you when choosing a lip balm? (e.g., natural ingredients, packaging, price)
Problems with Current Lip Balms	What don't you like about the lip balms you've tried?

Step 2: Write Your Design Brief

(State your personal need and what your lip balm will do)

I want to design a lip balm for myself because **[describe your lip care need or problem]**. My lip balm will be **[describe key features or qualities you want, e.g., natural, moisturizing, scented or unscented, eco-friendly packaging]** to help me **[what you want the lip balm to do or improve]**.

Personal Persona & Design Brief

Step 1: Create Your Personal Persona

Name	
What do you do most days? (e.g., sports, study, outdoor time)	
How often do you use lip balm? What do you like or dislike about the lip balms you've used?	
Do you have dry, sensitive, or normal lips? Do you prefer scented or unscented?	
What's important to you when choosing a lip balm? (e.g., natural ingredients, packaging, price)	
What don't you like about the lip balms you've tried?	

Step 2: Write Your Design Brief

(State your personal need and what your lip balm will do)

I want to design a lip balm for myself because **[describe your lip care need or problem]**. My lip balm will be **[describe key features or qualities you want, e.g., natural, moisturizing, scented or unscented, eco-friendly packaging]** to help me **[what you want the lip balm to do or improve]**.

PART 1: STUDENT GUIDANCE

How to Write Design Specifications (Criteria B, Strand 1)

What is a Design Specification?

A **design specification** is a list of **conditions and requirements** that your final product must meet in order to be successful.

Design specifications are **not design ideas**.
They do **not** describe exactly what your product will look like, smell like, or feel like.

Instead, they describe:

- What the product **must do**
- Who the product is **for**
- What the product **must avoid**
- What limits the product must **stay within**

You can think of design specifications as **rules** for your design.

What Design Specifications Are NOT

A design specification is **NOT**:

- “My lip balm will be vanilla scented”
- “My lip balm will be soft and creamy”
- “My lip balm will be pink”

These are **design choices**, and they come later.

At this stage, you are deciding **boundaries**, not **final decisions**.

The Golden Rule

> **Design specifications describe constraints, not creative choices.**

The Main Categories of Design Specifications

You will write your specifications under the following categories.
You must write at least one clear specification for each category.

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1. Functional Requirements

(What must the product do?)

These describe what the lip balm must be able to do when used.

Examples:

- The lip balm must moisturise dry lips.
- The lip balm must hold its shape at room temperature.
- The lip balm must be easy to apply without causing discomfort.

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2. User and Safety Requirements

(Who is it for and what must be avoided?)

These focus on comfort, daily use, and safety.

Examples:

- The lip balm must be suitable for daily use.
- The lip balm must be gentle enough for sensitive lips.

- The lip balm must not cause irritation when applied.

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3. Sensory Boundaries (Not Sensory Choices)

(Setting limits, not deciding features)

These describe what the product should **not** be like.

Examples:

- The scent must not be overpowering.
- The texture must not feel uncomfortable on the lips.
- The colour must be appropriate for everyday use.

Notice that no specific scent, texture, or colour is chosen.

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4. Ingredient and Material Constraints

(What must or must not be used?)

Examples:

- The lip balm must use skin-safe ingredients.
- The lip balm must avoid harmful or unsafe substances.
- The ingredients must be suitable for cosmetic use.

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5. Sustainability and Ethics

(How responsible is the product?)

Examples:

- The product should minimise waste.
- The ingredients should be responsibly sourced where possible.
- The product should avoid unnecessary environmental harm.

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6. Practical Constraints

(Real-world limits)

Examples:

- The lip balm must be possible to make safely in school.
- The product must be completed within the given time.
- The design must use the materials provided.

Writing Good Specifications: Sentence Starters

Students may use:

- “The lip balm must…”
- “The lip balm should…”
- “The product must not…”

Avoid:

- “I want…”
- “I like…”
- “I will choose…”

PART 2: TESTING CRITERIA

How Will You Know If Your Design Meets the Specification?

What Are Testing Criteria?

Testing criteria explain **how you will check** whether your design specification has been met.

If design specifications are the **rules**, testing criteria are the **checks**.

Every important specification should have **at least one matching test**.

Key Question to Ask

> “How can I check this without guessing?”

Testing does **not** need special equipment.
It can include:

- Observation
- Touch
- Simple user feedback
- Visual checks

Examples of Design Specifications and Testing Criteria

Example 1: Function

Design specification:
The lip balm must hold its shape at room temperature.

Testing criteria:
Leave the lip balm at room temperature for a set time and observe whether it keeps its shape without melting.

Example 2: User Comfort

Design specification:
The lip balm must be comfortable to apply on the lips.

Testing criteria:
Apply the lip balm once and check whether it spreads smoothly without dragging or causing discomfort.

Example 3: Sensory Boundary

Design specification:
The scent must not be overpowering.

Testing criteria:
Ask at least two users to smell the lip balm and note whether they find the scent mild or too strong.

Example 4: Practical Constraint

Design specification:
The product must be safe to make in school.

Testing criteria:
Check that all ingredients and steps follow the given safety guidelines and can be completed under supervision.

Important Reminder

Testing criteria do **not** explain how to improve the product.
They only explain **how to check** whether a specification is met.

PART 3: STUDENT TASK SHEET

Criteria B – Strand 1

Design Specification and Testing Criteria

Task Overview

Designers write design specifications to guide their ideas and decisions.
In this task, you will create a set of **design specifications** for your lip balm and explain **how each one will be tested**.

You are **not** choosing your final texture, scent, or colour yet.

PART A: Design Specifications

Complete the table below.

Category	Design Specification
Functional requirement	
User & safety requirement	
Sensory boundary	
Ingredient / material constraint	
Sustainability / ethics	
Practical constraint	

PART B: Testing Criteria

For each design specification above, explain **how you will test or check it**.

Design Specification	How will I test this?

PART C: Reflection (Answer in 4–5 sentences)

1. Why is it important to write design specifications before choosing final design ideas?
2. How do testing criteria help designers make fair and clear decisions?

Design Specifications & Testing Criteria – Lip Balm (MYP 1)

How can we clearly describe what our product must do and how we will test it?

Writing Design Specifications

Think about:

- Who is the product for?
- What must the product do?
- What must the product avoid?
- What limits the product?

Sentence Starters (Choose from below):

- The lip balm must ...
- The lip balm should ...
- The product must not ...
- I want the lip balm to...
- I would like the product to...

PART A: Design Specifications

Functional requirement	
User & safety requirement	
Sensory boundary	
Ingredient / material constraint	
Sustainability / ethics	
Practical constraint	

PART B: Testing Criteria

Design Specification	How will I test this?

Reminder: A good test can be repeated by someone else.

Writing Testing Criteria

Ask yourself:

- How can I check this without guessing?
- What can I see, feel, or observe?
- Who can give feedback?

Ways to Test (Choose from):

- Observation
- Visual check
- Touch test
- Simple user feedback
- Comfort check
- Sensory boundary check
- Practical constraint check

Task sheet

HELP BOX: Writing Design Specifications

Think about:

- Who is the product for?
- What must the product do?
- What must the product avoid?
- What limits the product?
- What boundaries must it stay within?

Sentence Starters (Choose from below):

- The lip balm **must** …
- The lip balm **should** …
- The product **must not** …
- I want the lip balm to …
- I would like the product to …

DESIGN SPECIFICATIONS

Functional requirement	
User & safety requirement	
Sensory boundary	
Ingredient / material constraint	
Sustainability / ethics	
Practical constraint	

HELP BOX: Writing Testing Criteria

Ask yourself:

- **How can I check this without guessing?**
- **What can I see, feel, or observe?**
- **Who can give feedback?**

Ways to Test (Choose from)

- **Observation**
- **Visual check**
- **Touch test**
- **Simple user feedback**
- **Comfort check**
- **Sensory boundary check**
- **Practical constraint check**

TESTING CRITERIA

Design Specification	How will I test this?

Teacher’s Notes (Not to be taught to Students)

1. Matter Begins with Atoms — But Materials Begin with Bonds

At the most fundamental level, all matter is composed of atoms. Every solid object, liquid, and gas around us is built from these microscopic particles. However, when we examine the observable properties of materials—such as hardness, brittleness, melting point, flexibility, gloss, viscosity, or spreadability—it becomes clear that atoms in isolation do not explain these behaviours. A single sodium atom does not possess the crystalline rigidity of table salt. A single carbon atom does not display the extraordinary hardness of diamond. The macroscopic properties that we associate with materials emerge only when atoms interact and organise through chemical bonding.

It is therefore conceptually incomplete to teach atomic structure without proceeding toward bonding and structural arrangement. The true explanatory pathway in materials science moves progressively from atoms to bonds, from bonds to structural organisation, and from structure to observable properties. This structural thinking is central not only to chemistry but also to material design and engineering. Once teachers internalise this continuum, they are able to guide students beyond memorisation toward causal reasoning.

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2. Why Do Atoms Form Bonds?

Atoms form bonds because bonded arrangements are energetically more stable than isolated atoms. Stability in chemical systems is governed by energy minimisation. When atoms bond, the total potential energy of the system decreases relative to the energy of the separated atoms. This lowering of energy creates a more stable state.

There are two primary ways by which atoms achieve this stabilisation. One mechanism involves the transfer of electrons from one atom to another, resulting in oppositely charged ions that attract each other. The other mechanism involves the sharing of electrons between atoms so that both achieve stable electronic configurations. These two processes correspond to ionic bonding and covalent bonding respectively. The nature of electron interaction determines the strength and type of attraction between particles, and this attraction ultimately governs how those particles arrange themselves in space. The arrangement, in turn, determines material behaviour.

3. Ionic Bonding and Electrostatic Structural Solids

Ionic bonding occurs when one atom transfers one or more electrons to another atom. This process typically occurs between metals, which readily lose electrons, and non-metals, which readily gain electrons. When sodium transfers one electron to chlorine, sodium becomes a positively charged ion (Na^+) and chlorine becomes a negatively charged ion (Cl^-). These oppositely charged ions attract each other strongly through electrostatic forces.

It is important to emphasise that ionic compounds do not exist as isolated pairs of ions forming discrete molecules. Instead, each ion interacts simultaneously with multiple oppositely charged neighbours. The result is an extended three-dimensional arrangement known as a crystal lattice. In sodium chloride, each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions. This pattern repeats uniformly in all spatial directions, producing a highly ordered structure.

The repeating lattice arrangement minimises repulsive interactions and maximises attractive interactions, leading to structural stability. Because electrostatic forces between ions are strong and act throughout the entire lattice, ionic solids typically exhibit high melting points and considerable rigidity. However, they are also brittle. Brittleness arises because when layers of ions are forced to shift relative to one another, like charges may align adjacent to each other. This alignment produces strong repulsive forces, causing the crystal to fracture rather than deform. Thus, the mechanical behaviour of ionic solids can be directly explained by their lattice structure and bonding type.

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4. Covalent Bonding and Shared Electron Systems

Covalent bonding occurs when atoms share pairs of electrons to achieve stable electronic configurations. Unlike ionic bonding, which relies on electrostatic attraction between oppositely charged ions, covalent bonding involves directional sharing of electron density between specific atoms. However, covalent bonding can produce very different types of solid structures depending on how extensively the sharing extends through the material.

One class of covalent structures is the giant covalent network. In these materials, covalent bonds extend continuously throughout the entire solid. Diamond is a classic example. In diamond, each carbon atom forms four strong covalent bonds arranged tetrahedrally. These bonds connect to other carbon atoms, and the bonding pattern extends in three dimensions without forming separate molecules. Because the entire structure is held together by strong covalent bonds, diamond exhibits exceptional hardness and an extremely high melting point. The rigidity of diamond arises not simply from the strength of individual bonds but from the continuous network that distributes mechanical stress throughout the structure.

In contrast, simple molecular solids consist of discrete molecules in which covalent bonds exist only within each molecule, not between neighbouring molecules. The molecules themselves are held together by comparatively weak intermolecular forces such as van der Waals interactions. Wax is an example of such a substance. Within each wax molecule, atoms are strongly bonded covalently. However, between molecules, only weak attractions operate. When wax cools from a molten state, its molecules may arrange into ordered regions, forming crystalline domains. Yet the forces holding these domains together are weak compared to the forces in ionic or giant covalent solids. Consequently, molecular solids generally have lower melting points and are softer and more deformable.

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5. The Meaning of Crystallinity

A crystalline structure is characterised by a repeating, ordered arrangement of particles extending over long distances. Crystallinity refers to structural order rather than mechanical strength. The misconception that all crystals must be hard arises from overgeneralisation based on ionic and giant covalent examples. In reality, the strength of a crystalline solid depends primarily on the nature of the bonding that holds the particles together.

In ionic crystals, strong electrostatic interactions produce rigid, high-melting solids. In giant covalent crystals, continuous covalent bonding produces exceptional hardness. In molecular crystals, however, the repeating arrangement is maintained only by weak intermolecular forces, resulting in softer materials. Thus, crystallinity describes spatial organisation, while bonding type determines mechanical behaviour. It is the combination of both that defines material properties.

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6. From Crystalline Solids to Soft Materials

Not all crystalline systems behave as rigid solids. Some materials exhibit partial crystallinity interspersed with less ordered regions. Waxes belong to this category. When molten wax cools, molecular motion decreases and ordered packing begins to occur. Small crystalline regions form and interlock with one another, creating a three-dimensional network. However, because the interactions between wax molecules are intermolecular rather than ionic or network covalent, the resulting structure is comparatively weak.

The wax network is strong enough to hold its shape at room temperature, yet weak enough to deform under applied pressure. This dual behaviour—shape retention combined with deformability—distinguishes such materials from both brittle crystals and flowing liquids. It represents an intermediate state governed by structural organisation rather than by a simple classification of solid versus liquid.

---

7. Semi-Solid Colloidal Systems

To understand semi-solid systems fully, one must introduce the concept of a colloid. A colloid is a system in which one phase is dispersed within another continuous phase without forming a true solution. The dispersed particles are small enough to remain distributed throughout the system but large enough to influence its mechanical and optical properties. Colloidal systems are structured dispersions rather than homogeneous solutions.

When the continuous phase is solid-like and the dispersed phase is liquid, the result is a semi-solid colloidal system. In such systems, the internal network provides mechanical stability while the dispersed liquid modifies flow behaviour. Semi-solids maintain their shape under normal conditions but deform when stress is applied. They do not pour freely like liquids, nor do they fracture immediately like brittle solids. Their behaviour emerges from the interplay between structural network formation and internal dispersion.

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8. The Crystalline Wax Network in Lip Balm

Lip balm provides a practical example of a semi-solid colloidal system. At a technical level, it consists of a crystalline wax network that traps liquid oils within its structure. When wax cools during formulation, its molecules crystallise and interlock, forming a three-dimensional skeleton. This skeleton establishes mechanical integrity and prevents the oils from flowing freely. Unlike ionic lattices, however, the wax network is maintained by weak intermolecular forces. Therefore, it remains flexible rather than rigid.

The oils in the formulation occupy the spaces between wax crystals. They do not crystallise at room temperature and thus function as the dispersed liquid phase. These oils act as plasticisers, reducing brittleness and allowing the material to deform smoothly under pressure. When lip balm is applied, the network partially yields, releasing oil to the surface and enabling spreadability. Once the pressure is removed, the structure largely retains its form.

Because the system is structurally integrated, altering the proportion of wax or oil changes the entire behaviour of the material. Increasing wax concentration increases crystalline density, strengthening the internal network and raising firmness while reducing spreadability. Increasing oil concentration disrupts network continuity, enhancing glide but lowering mechanical resistance. Additives such as pigments occupy physical space within the matrix and can interfere with crystal packing, thereby modifying tactile perception. Volatile scent components dissolve in the oil phase and may subtly influence melting behaviour and sensory response.

Thus, a semi-solid colloidal system must be understood as an integrated structural arrangement rather than as a simple mixture of ingredients. Its properties emerge from the balance between crystalline organisation and dispersed liquid phases. This understanding completes the conceptual journey from atomic bonding to macroscopic behaviour.

1. WHAT A LIP BALM IS (TECHNICAL VIEW)

At a technical level, a lip balm is:

> **A semi-solid colloidal system composed of a solid lipid network that traps liquid oils and dispersed additives.**

Let's unpack this slowly.

2. WHAT IS A SEMI-SOLID COLLOIDAL SYSTEM?

A. Start with "colloid" (core idea)

A **colloid** is a mixture where:

- One substance is **evenly dispersed**
- Inside another substance
- Without fully dissolving

Key feature:

> The particles are **too small to see**, but **large enough to affect texture**.

B. Everyday colloid examples (important for teaching)

Product	Colloidal system type	Why it matters
Milk	Liquid-in-liquid	Fat droplets in water
Butter	Water-in-fat	Semi-solid, spreadable
Jelly	Liquid-in-solid	Holds shape but jiggles
Cream	Oil-in-water	Smooth, spreadable
Lip balm	Liquid-in-solid	Semi-solid, stable

So lip balm behaves more like **butter or waxy jelly**, not liquid oil.

C. Why “semi-solid”?

A **semi-solid**:

- Holds its shape at room temperature
- Deforms under pressure (your finger or lips)
- Returns partially to shape

This is why:

- Lip balm doesn’t pour
- But still spreads

D. Lip balm’s specific colloidal structure

Technically, lip balm is:

> **Liquid oils dispersed within a crystalline wax network**

Think of it like:

- A **3D wax skeleton**
- With oil **trapped inside**

This structure controls:

- Firmness
- Melting point
- Spreadability

---

3. ROLE OF EACH INGREDIENT (ADVANCED FUNCTION)

1. WAX — STRUCTURAL MATRIX (CRYSTALLINE NETWORK)

Scientific role

- Waxes crystallize as they cool
- They form a **rigid lattice**
- This lattice traps oils

Key property

- Melting point (typically 60–70°C)

What wax controls

- Shape stability
- Heat resistance
- Mechanical strength

Too much wax

- Dense crystal network
- Balm feels hard
- Poor oil release → drag on lips

Too little wax

- Weak network
- Oil leaks out

- Balm melts easily

Design takeaway

Wax percentage controls **solid strength** of the colloid.

---

2. OILS / BUTTERS — DISPERSED LIQUID PHASE

Scientific role

- Plasticise the wax network
- Reduce brittleness
- Provide lubrication

What oils actually do

- Sit between wax crystals
- Allow movement under pressure
- Provide glide and moisture

More oil

- Softer matrix
- Increased gloss
- Faster oil migration to lips

Less oil

- Stiffer balm
- Reduced lubrication
- Increased friction

Design takeaway

Oils control **flow under pressure**.

---

3. PIGMENT — SOLID PARTICULATE DISPERSION

Pigments are:

- Insoluble solid particles
- Suspended in oil
- NOT dissolved

This is crucial.

Scientific role

- Light absorption and reflection
- Visual colour intensity

What pigment affects

- Opacity
- Colour payoff
- Mouthfeel (if overloaded)

Higher pigment concentration

- More particles competing for space
- Increases friction
- Can feel gritty if poorly dispersed

Lower pigment concentration

- Transparent or sheer look
- Smoother texture

Design takeaway

Pigment affects both **appearance AND tactile feel**.

---

4. SCENT (ESSENTIAL OILS) — VOLATILE FUNCTIONAL ADDITIVE

Scientific role

- Volatile compounds evaporate easily
- Reach olfactory receptors

Important technical point

- Essential oils act like **active chemicals**
- Not just “smell”

Higher scent concentration

- Faster evaporation
- Strong initial smell
- Increased irritation risk

Lower scent concentration

- Subtle aroma
- Longer usability
- Safer for sensitive users

Design takeaway

Scent is **dose-sensitive**, not decorative.

---

4. WHY PROPORTION CHANGES EVERYTHING

A. Colloidal balance principle

In a colloidal system:

Changing the amount of ONE component changes the ENTIRE structure.

Example:

- More pigment = less space for oil
- More oil = weaker wax network
- More scent = altered oil phase behaviour

This is **systems thinking**, not ingredients thinking.

B. Pigment proportion: what really happens

Pigment level	System effect	User experience
None	Clear oil-wax matrix	Smooth, invisible
Low	Light scattering	Natural tint
High	Particle crowding	Strong colour, heavier feel

Key idea:

Pigment competes with oil for space in the matrix.

C. Scent proportion: volatility effect

Essential oils:

- Are liquid
- But evaporate
- Change oil behaviour

More scent:

- Lowers overall melting point slightly
- Can soften balm
- Increases sensitivity risk

Less scent:

- Stable structure
- Neutral feel

---

5. HOW YOU TRANSLATE THIS TO STUDENTS (SIMPLIFIED)

What YOU know:

- Crystalline networks
- Dispersed phases
- Volatile compounds

What STUDENTS hear:

- “Wax holds everything together”
- “Oil makes it smooth”

- “Too much colour or smell can cause problems”
- “Balance matters”

You are teaching **design cause–effect**, not chemistry terms.

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6. CLASSROOM EXAMPLE YOU CAN USE CONFIDENTLY

Analogy: Brick Wall

- Wax = bricks
- Oil = mortar
- Pigment = coloured stones
- Scent = air freshener

Too many bricks → wall is rigid
Too much mortar → wall collapses
Too many stones → uneven wall
Too much air freshener → overpowering smell

This analogy is technically accurate enough.

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7. WHY THIS MATTERS FOR IB DESIGN

This depth supports:

- **B.i**: Writing precise specifications (“light tint”, “mild scent”)
- **B.ii**: Sensory design sketches
- **B.iii**: Evidence-based justification
- **Criterion D**: Evaluation of success

DETAILED FUNCTION OF EACH INGREDIENT CATEGORY

4.1 Beeswax — STRUCTURAL MATRIX (ESSENTIAL)

Material role

Beeswax is the **primary structural agent**.

On cooling:

- Beeswax molecules crystallise
- Crystals interlock
- A rigid but flexible network forms

This network:

- Holds oils in place
- Prevents leakage
- Controls melting behaviour

Design control

Beeswax determines:

- Firmness
- Heat resistance
- Shape retention

Design consequences

- **High beeswax** → hard, draggy balm

- **Low beeswax** → soft, unstable balm
- **No beeswax** → liquid oil product

Design insight

> Beeswax defines the boundary between a balm and an oil.

4.2 Oils — MOISTURE & FLOW (ESSENTIAL)

Oils form the **continuous phase** of the balm.

They are the part that:

- Touches the lips
- Provides lubrication
- Delivers conditioning

Coconut Oil (Cold-Pressed)

- Light, fast-spreading
- Temperature sensitive
- Softens the balm

Castor Oil

- Thick and viscous
- High gloss
- Excellent pigment suspension

System behaviour

Oils:

- Sit between wax crystals
- Allow movement under pressure
- Determine spreadability

Design consequences

- More oil → softer, glossier balm
- Too much oil → weak structure

4.3 Shea Butter — SEMI-SOLID MODIFIER (OPTIONAL)

Shea butter is:

- Solid at room temperature
- Liquid at skin temperature

This places it **between wax and oil**.

Function

- Softens the wax lattice
- Adds creaminess
- Improves comfort

Critical distinction

Shea butter:

- Does **not** create structure alone

- Does **not** replace oil
- Does **not** replace wax

Design role

Shea butter fine-tunes feel, not form.

4.4 Pigments — SOLID PARTICLE DISPERSION (OPTIONAL)

Pigments are:

- Insoluble solid particles
- Dispersed in oil
- Suspended, not dissolved

Natural pigments used

- Beetroot powder
- Alkanet root powder
- Hibiscus powder
- Cocoa powder

System impact

Pigments:

- Occupy physical space
- Interact with oils
- Increase friction if overloaded

Design consequences

- Light pigment → sheer tint, smooth feel
- Heavy pigment → strong colour, heavier mouthfeel

Critical design truth

> Pigment improves appearance, not care.

4.5 Scents & Flavour Oils — VOLATILE ADDITIVES (OPTIONAL)

Scents are:

- Volatile compounds
- Dissolved in oil
- Evaporate over time

Essential oils

- Peppermint → strong, cooling, irritant-prone
- Sweet orange → lighter, uplifting, fades faster

Flavour oils

- Strawberry, vanilla
- More stable
- Cosmetic rather than therapeutic

System behaviour

Scents:

- Do not affect structure
- Alter user experience dramatically
- Can cause discomfort if excessive

Design implication

> Scent must be designed for inclusivity, not intensity.

4.6 Vitamin E — STABILITY & CONDITIONING (FUNCTIONAL ADDITIVE)

Vitamin E (tocopherol):

- Acts as an antioxidant
- Slows oil oxidation
- Extends shelf life

It:

- Does not affect firmness
- Does not affect colour
- Does not affect scent intensity significantly

Design role

> Vitamin E protects the system over time.

Lesson 1 and 2:

Why atoms bond + Ionic bonding + Covalent bonding + Compare structures

Video Video

Quiz link

Lesson 3:

Crystal lattice formation + brittleness vs hardness

Video

Lesson 4:

Colloid vs Semi colloid and connection to lip balm.

Video

Padlet reflection - Connection to SEL

The video links shared here are not to be discussed completely. However, they are just for guidance.



Video

Part A: What Is Essential? (Non-Negotiables)

These are the things a lip balm **must have** to work as a lip balm.

Task:
List the essentials and explain *why each one is necessary*.

Use complete sentences.

Guiding questions (do not answer all, use them to think):

- What ingredients are required for lip balm to exist as a semi-solid?
- What gives the balm its structure?
- What allows it to moisturise and spread?
- What would happen if this was missing?

Student response space:

I think the essential parts of a lip balm are…
This is essential because…

Part B: What Makes a Lip Balm Better? (Quality Enhancers)

These are features that are **not required**, but improve comfort, experience, or usability.

Task:
Identify features or ingredients that improve the lip balm and explain *how* they improve it.

Guiding questions:

- What makes the balm feel smoother or creamier?
- What improves how pleasant it is to use?
- What helps the product last longer or feel more comfortable?

Student response space:

Some things that make a lip balm better are…
These improve the product because…

Part C: What Needs Careful Discussion? (Design Decisions)

Some choices are **not clearly right or wrong**. These require designers to think about users, sensitivity, balance, and ethics.

Task:
Identify aspects of lip balm design that need discussion and explain *why they are debatable*.

Guiding questions:

- Can too much of a good thing become a problem?
- Do all users want the same colour or scent?
- How can pigment or scent affect comfort?
- Why does balance matter in a semi-solid system?

Student response space:

Some design choices need discussion, such as…
These need careful thinking because…

Part D: Final Reflection (Thinking Like a Designer)

Answer in 4–6 sentences.

Prompt:

After learning how lip balm works as a semi-solid system, explain why good design is about balance rather than adding more ingredients.

The statement was:

A semi-solid colloidal system is a colloid where the continuous phase forms a soft structural network.

To understand this deeply, we must clarify three ideas:

1. What is a continuous phase?
2. What is a structural network?
3. Why does that network make the material semi-solid?

First, consider what a colloid is at the particle level.

In any colloidal system, there are always two parts. One substance is spread throughout another. The substance that is spread out is called the dispersed phase. The substance that surrounds it and extends throughout the entire material is called the continuous phase (or dispersion medium).

The continuous phase is important because it is the part that connects everything. It forms the “background” material that holds the system together.

Now let us move to the phrase “forms a soft structural network.”

In some colloids, the continuous phase is simply a liquid. For example, in milk, water is the continuous phase and tiny fat droplets are dispersed in it. Milk still behaves like a liquid because the continuous phase is liquid and does not form any rigid structure.

But in a semi-solid colloidal system, something different happens.

The continuous phase does not remain a simple fluid. Instead, its particles arrange themselves into an interconnected structure — a network. This network may form because:

- Molecules crystallise and interlock (as in wax),
- Long polymer chains entangle,
- Or weak intermolecular forces create an extended framework.

This network spreads throughout the entire material. It acts like a three-dimensional scaffold.

However, the network is not rigid like an ionic crystal lattice. The forces holding it together are weaker — often intermolecular forces rather than strong ionic or giant covalent bonds. Because of this, the network is soft and flexible.

Now consider what this means mechanically.

Since the continuous phase forms a connected network:

- The material can hold its shape.
- The dispersed phase becomes trapped inside this network.
- The system does not flow freely.

But because the network is soft:

- It can deform when pressure is applied.
- It can bend rather than fracture.
- It can release part of the dispersed phase under stress.

This combination produces semi-solid behaviour.

Let us apply this to lip balm to make it concrete.

In lip balm:

- Wax molecules cool and form small crystalline regions.
- These crystalline regions interlock.
- Together, they create a three-dimensional network.

That network is the continuous phase.

Liquid oil is trapped within that network. The oil is the dispersed phase.

Because the wax network spans the entire balm, the balm holds its shape. But because the network is made of weak intermolecular forces, it is soft enough to deform when you press it against your lips. The oil then spreads out.

So when we say:

“A semi-solid colloidal system is a colloid where the continuous phase forms a soft structural network,”

we mean:

The material contains two phases, but the phase that surrounds everything has organised itself into a connected, flexible structure that gives the whole system its shape and mechanical behaviour.

In contrast:

If the continuous phase were fully liquid and unstructured, the material would flow like a solution.

If the network were rigid and held by strong ionic or giant covalent bonds, the material would behave like a hard

crystal.

A semi-solid lies in between because the internal network is structured but soft.

Link for tasksheet

PART 1 – TEXTURE PROFILES

How might the balm behave when someone uses it?

Create **at least THREE** different texture possibilities.

Think about:

- How easily it spreads
- How it feels during use
- What it feels like after
- When it would be useful

Texture Idea 1

Spreadability:
(very easy / moderate / difficult)

Feeling during use:

After-feel:

Who would like this:

When it is useful:

Possible problem:

Texture Idea 2

Spreadability:

Feeling during use:

After-feel:

Who would like this:

When it is useful:

Possible problem:

Texture Idea 3

Spreadability:

Feeling during use:

After-feel:

Who would like this:

When it is useful:

Possible problem:

PART 2 – FORMULATION CONCEPT IDEAS

What type of ingredient balance might create these behaviours?

You are NOT writing amounts.

You are deciding:

- which part of the system is stronger
- which part is lighter

Examples:

- oil dominant
- structure dominant
- balanced

Concept A

Balance idea:

What texture might this create:

Why this might suit the user:

What might be difficult:

Concept B

Balance idea:

What texture might this create:

Why this might suit the user:

What might be difficult:

Concept C

Balance idea:

What texture might this create:

Why this might suit the user:

What might be difficult:

PART 3 – SCENT EXPERIENCE OPTIONS

How might the product feel emotionally?

You must explore **different possibilities**.

Option	Mood created	Who might enjoy it	Who might avoid it	Possible issue
Option 1				
Option 2				
Option 3				

Stage	Activity	Student Task	Deliverable	ATL Skills
1. Inspiration Wall	Show examples of branded lip balm tubes (Burt's Bees, Eos, etc.)	Note colors, fonts, label styles, eco-labels	Observation sheet or sticky notes	Media literacy, creative thinking
2. Design Prompt	"What kind of feeling should your lip balm give the user?"	Choose theme (e.g., playful, soothing, elegant) and brainstorm scent, color, brand personality	Word cloud / mood board	Creative thinking
3. Sketching	Draw 2–3 *label concepts* for the tube	Include:- Name & logo- Color palette- Icons (cruelty-free, organic)- Key info layout	Sketches + label annotations	Communication, creativity
4. Peer Review	Gallery walk with "Star & Wish" feedback	Refine designs	Feedback notes	Collaboration

Step	What You Do (Teacher)	What Students Do	ATL Skill Focus
1. Review Design Options	Ask students to revisit their 2–3 design sketches and feedback.	Compare and evaluate design ideas using a decision-making matrix or SWOT table.	**Critical thinking**
2. Choose Final Design	Guide students in selecting their best idea based on: feasibility, sustainability, aesthetics, and function.	Select the strongest design and briefly justify the choice (50–100 words).	**Decision-making**
3. Create Canva Mockup	Provide Canva access + optional mockup template (tube, background).	Recreate their **final design** digitally on Canva: include product name, label, color, scent, icons, and packaging visuals.	**Media literacy, creativity**
4. Add Annotations	Model or show an example of an annotated visual.	Add clear **annotations** to their design mockup explaining:– Why they chose this color, scent, font– What the icons/symbols represent– Sustainability or ethical choices	**Communication**
5. Submit Final Design Document	Share submission format (PDF or slideshow).	Submit a final digital page/slideshow with: ✔️ Canva mockup ✔️ Annotations ✔️ Short written rationale (100–150 words)	**Organization**

Stage	Activity	Student Task	Deliverable	ATL Skills
1. Ingredient Selection	Review ingredient sheet	Choose: wax, oils, scent, color (if any)	Ingredient checklist	Research, organization
2. Recipe Development	Follow a recipe planner template	Include:- Ingredient quantities- Tools- Step-by-step procedure- Tube filling method	Recipe flowchart or worksheet	Time management
3. Safety & Hygiene	Watch demo or read safety card	Complete safety checklist	Checklist	Responsibility
4. Label Printing Plan	Decide final label dimensions for tubes	Create digital version or outline printing steps	Label mockup or file	Technical literacy

Task: Choose your ingredients:

- Wax: _____
- Oils: _____
- Scent: _____
- Color (optional): _____

Recipe Development

- **Instructions:** Use the Recipe Planner Template.
- **Task:** Complete the following:
    - Ingredient Quantities:
        1. Wax: _____
        2. Oil: _____
        3. Scent/Color: _____
    - Tools Needed: _____
    - Step-by-step Procedure:
        1. _____
        2. _____
        3. _____

- Tube Filling Method: _____

Safety & Hygiene

Task: Create a checklist.

FORMULATION PLANNING & RECIPE DEVELOPMENT

SECTION 1 - DETAILED PLANNING & INGREDIENT SELECTION

Ingredient Selection & Functional Notes

Component	Selected Ingredient(s)	Function in the lip balm
Wax		
Oil(s)		
Butter (if used)		
Scent		
Colour (optional)		
Vitamin E (if used)		

SECTION 2 – QUANTITY PLANNING & PROCEDURE

Ingredient	Planned Quantity
Wax	
Oil(s)	
Butter	
Scent	
Colour	
Vitamin E	
Total Weight	

Important: They must justify amounts in relation to texture and finish.

Tools Required

Tool	Purpose
1.	
2.	
3.	

C. Step-by-Step Procedure

Step No.	Action	Why this step is important
1		
2		
3		
4		
5		

Tube Filling Plan

Question	Response
At what stage will you pour the mixture?	
How will you prevent air bubbles?	
How long will cooling take?	
How will you check final texture?	

Safety & Hygiene Procedure

Safety Action	How will you?
Work surface cleaning	
Hygiene	
Tools sanitised	
Heat handled safely	

Ingredients measured accurately	
Containers sealed properly	

Additional safety consideration: _____

Explicit Teaching of the procedure

Step No.	Production Step	Resources Needed	Time Needed (mins)	Notes / Safety Tips
1	Infuse color powder in oil	Natural powder (e.g., beetroot), carrier oil, double boiler		Stir gently; heat on low; strain before use
2	Set up clean workspace	Table cover, gloves, sanitizer, paper towel		Wash hands and sanitize surface
3	Gather & measure ingredients	Shea butter, beeswax, infused oil, digital scale		Be accurate; avoid cross-contamination
4	Melt & mix base ingredients	Double boiler, heat-safe bowl, spatula, oven mitts		Supervise heat use; melt slowly
5	Add infused oil (with color/scent)	Strained infused oil, essential oil, dropper		Add after removing from heat; stir well
6	Pour mixture into lip balm tubes	Pipette or funnel, lip balm tubes		Fill quickly before mixture solidifies
7	Let cool and solidify	Cooling rack or fridge space		Do not move tubes during cooling
8	Label and decorate	Labels, scissors, glue/tape, Canva printout		Ensure label size fits tube neatly
9	Clean up & store product	Cleaning cloths, soap, water		Wipe all surfaces and wash equipment thoroughly

Lip Balm Recipe Card - Students make the recipe card, once the teacher explains the procedure and quantities -To be assessed as formative

Quantity: 1 Tube (~5g)

Ingredients:

- Beeswax: **1.25 g**
- Shea or Cocoa Butter: **1.25 g**
- Carrier Oil (coconut, almond, etc.): **2 g**
- Infused Oil (optional): **0.5 – 1 g**
- Essential Oil (optional): **1–2 drops**
- Natural Colorant (optional): **tiny pinch**

Equipment:

- Heatproof container or double boiler
- Stirring stick or spoon
- Precision scale (0.01 g accuracy)
- Lip balm tube (5 g size)
- Measuring spoons (for drops)

Procedure:

Step 1: Prepare Infused Oil (Day Before)

- Mix your chosen natural powder with carrier oil in a small jar.

- Let it infuse overnight at room temperature.

Step 2: Melt Beeswax & Butter

- Use a double boiler or heatproof container over simmering water.
- Stir occasionally until fully melted.

Step 3: Add Oils

- Remove from heat.
- Stir in carrier oil and infused oil until well combined.

Step 4: Add Essential Oils & Color

- Add 1–2 drops of essential oil and optional natural colorant.
- Mix thoroughly.

Step 5: Fill Tubes

- Pour liquid balm into tubes promptly before solidifying.

Step 6: Cool & Set

- Leave tubes at room temperature or refrigerate to set.

Step 7: Label & Store

- Label tubes with scent, color, and date.
- Store in a cool, dry place.

Tips & Safety:

- Adjust beeswax quantity for desired firmness.
- Stir constantly for even mix.
- Test essential oils for allergies before use.
- Handle hot mixtures with care to avoid burns.
- Use ethically sourced, natural ingredients when possible.

Production Passport: My Lip Balm Diary

1. What did I make today?

Write the name of your lip balm and describe it in 2–3 sentences. Think about its color, scent, texture, and who it's for.

Example:
I made a light pink, vanilla-scented lip balm with a smooth texture, perfect for people with dry lips.

My entry:

2. What ingredients and materials did I use?

List everything you used, including amounts. Where did you source your ingredients?

My list:

3. What tools did I use?

Write down the tools and equipment you needed (scales, double boiler, stirring sticks, tubes, etc.)

My tools:

4. How did I make it?

Describe the steps you followed to make your lip balm. Remember to include making the infused oil the day before! You can use bullet points or short paragraphs.

My process:

5. How long did each step take?

Estimate the time taken for each part of the process (melting, mixing, filling, setting, etc.)

Example:
Melting – 10 mins
Mixing – 5 mins

My timings:

6. What challenges did I face?

Write about any problems you encountered and how you solved them. Did anything surprise you?

My challenges & solutions:

7. How did I stay safe?

What safety precautions did you follow during the process?

My safety notes:

8. Changes to My Action Plan & Why

Did you change anything from your original plan? Explain what you changed and why you made these adjustments.

Changes & justification:

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9. My photo story

Attach photos or videos of your process here. Add a short caption for each.

Example captions:

- Measuring shea butter
- Pouring mixture into tubes

My photo captions:

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10. What did I learn and how do I feel about my product?

Reflect on your experience. What worked well? What would you improve next time? Did the final product meet your expectations?

My reflection:

Criteria D_Formative - **To be assessed as formative.**

Scaffolding Sheet

MYP1_Criteria D - **To be assessed as summative**

1. All these products failed because the designer did not *fully understand the user's real needs and context of use*.

Other acceptable variants:

- consider how and where the product would be used
- think about the user's comfort, environment, and habits
- test the product in real-life conditions

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2. A strong problem statement would have prevented this by *clearly identifying the user, the problem, and the reason the problem exists*.

Other acceptable variants:

- focusing on the real need instead of jumping to a solution
- defining the problem before designing the product
- highlighting constraints such as climate, usability, or sustainability

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3. Good designers must analyse *the user, the environment, and existing product limitations* before creating solutions.

Other acceptable variants:

- user needs and real-world conditions
- problems, causes, and consequences
- what is not working in current products

Quantity Plan for a 10 g Balm

Ingredient	Quantity
Beeswax	3 g
Coconut oil	4 g
Shea butter	2.8 g
Beetroot powder	0.2 g

Step-by-Step Demonstration

Step 1 — Measure Ingredients

Place the beaker on the digital scale.

Add:

- **3 g beeswax**
- **4 g coconut oil**
- **2.8 g shea butter**

Step 2 — Heat the Ingredients

Place the beaker in a **double boiler**.

Heat gently until everything melts.

- Beeswax melts around **62–65°C**
- Oils melt earlier

- Gentle heating prevents burning

Step 3 — Mix the Ingredients

Stir slowly with a glass rod.

Stirring helps distribute wax molecules evenly throughout the oils, which allows the balm to solidify uniformly.

Step 4 — Add Pigment Powder and Flavor oil

Add a **very small pinch (about 0.2 g)**

Stir thoroughly.

- Pigments must disperse evenly

- Too much pigment can make balm grainy
- Remove from heat and add flavor oil.

Step 5 — Pour Into Container

Pour the mixture carefully into the lip balm tube.

Pour while the mixture is still liquid but not extremely hot.

Step 6 — Cooling Phase

Leave the tube undisturbed.

Cooling time: **10–15 minutes**

As the mixture cools:

- Beeswax forms a **crystalline network**

- Oils become trapped inside this structure

- The liquid mixture becomes **semi-solid**

Equipment Needed

Tool	Purpose
Digital scale	Accurate measurement
Heatproof glass beaker	Mixing container
Stirring rod / spoon	Mixing
Double boiler or hot plate	Gentle heating
Lip balm tubes or small container	Final product
Pipette or dropper	Controlled pouring

LIP BALM CUSTOMIZATION PREFERENCE SHEET

Visitor Name (optional): _____
Date: _____

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1. Texture Preference (choose one)
[ ] Matte (no shine, natural look)
[ ] Semi-matte (slight shine, balanced feel)
[ ] Glossy (high shine, smooth finish)
[ ] Extra moisturizing (thick, rich feel)

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2. Flavor Preference (choose one)
[ ] Sweet Orange
[ ] Vanilla
[ ] Strawberry
[ ] No flavor

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3. Color Preference (choose one)
[ ] Dark Pink
[ ] Light Pink
[ ] Brown / Nude
[ ] No color (clear balm)

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4. Additional Preferences (optional)
Fragrance:
[ ] Strong
[ ] Mild
[ ] No fragrance

Feel:
[ ] Lightweight
[ ] Long-lasting / Heavy

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5. Any allergies or sensitivities?

6. Student Notes (for preparation)

7. Visitor Feedback (after use)

[ ] Loved it
[ ] Good
[ ] Can improve

Comments: